Group Audit Questions – Risk of AI Hallucination

1. **Coverage Gaps**

   - Does the dataset fully cover the topic or are there missing areas that could force the AI to guess?

   A: Not full coverage; lacks latest/global car models.

   - Are there rare or edge cases not represented?

   A: Rare or discontinued cars underrepresented.

2. **Data Clarity**

   - Is the data clearly labeled, or could ambiguous terms lead to misinterpretation by the AI?

   A: Structured datasets clear; Edmunds text ambiguous.

   - Are there multiple definitions for the same term (e.g., “slice” vs. “dice” in cooking)?

   A: Some terms overlap (e.g., “mileage,” “fuel efficiency”).

3. **Conflicting Information**

   - Do any sources contradict each other?

   A: Opinions vs. factual data may conflict.

   - How will the AI know which one to trust?

   A: AI should prioritize EPA/NHTSA as reliable sources.

4. **Outdated Information**

   - Is any data old enough to be inaccurate?

   A: Cooper Union (2017) outdated; others current.

   - Are there fast-changing facts in the dataset that could cause errors?

   A: Rapid changes in car prices and features may cause inaccuracies.

5. **Source Reliability**

   - Are the data sources trustworthy, or do they have a reputation for errors, bias, or exaggeration?

   A: EPA/NHTSA highly trustworthy; Kaggle depends on uploader.

   - Is any source opinion-based rather than fact-based?

   A Edmunds includes opinion-based reviews.

6. **Data Bias**

   - Does the dataset favor certain perspectives, geographies, or demographics, leaving the AI blind to others?

   A: U.S.-centric focus limits diversity. AI may ignore international models or markets.

7. **Overgeneralization Risk**

- Could small samples cause the AI to assume patterns that aren't true?

A: Limited EV or hybrid samples may lead AI to false assumptions about overall car trends.

8. **Context Loss**

- Are there data points that only make sense with context that's missing from the dataset?

A: Review tone and sarcasm may be misread without full text understanding.

9. **Inaccessible References**

- Does the dataset reference external materials (websites, PDFs, books) that the AI won't have direct access to?

A: Kaggle requires sign-in; not all linked sources are accessible to AI tools.

10. **Unverified Assumptions**

- Is any part of the dataset based on assumptions rather than evidence?

A: Relies on authenticity of user reviews and correctness of uploaded data.